

Physics-Infused Reduced-Order Modeling for Analysis of Multi-Layered Hypersonic Thermal Protection Systems With Unknown Thermal Contact Resistances

Abstract

This work presents a systematic design and implementation of the physics-infused reduced-order modeling (PIROM) framework for optimal inference and reduced-order modeling of hypersonic TPS with unknown thermal contact resistances (TCRs). The PIROM framework is designed to leverage the strengths of both physics-based and data-driven modeling paradigms, by infusing physics-based models with data-driven components to enhance their predictive capabilities and generalizability. Inference heat conduction problems (IHCPs) are solved using the PIROM framework to infer TCRs from internal temperature measurements, while providing accurate predictions of transient TPS temperature dynamics under varying boundary conditions and material properties. Particularly, experimental limitations are considered in the design of the PIROM framework, including noisy observables and thermocouple placement limitations. The performance of the PIROM framework is benchmarked against traditional physics-based modeling approaches (e.g., one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.

1 Introduction

What is the importance of hypersonic TPS? Hypersonic thermal protection systems (TPS) play a critical role in safeguarding hypersonic vehicles as they are exposed to extreme aerodynamic heating during flight. Therefore, accurate modeling and prediction of TPS behavior become essential for designing effective thermal protection strategies. Nonetheless, the highly non-linear fluid-thermal-structural interactions, temperature-dependent material property variations, and uncertainties in thermal contact resistances (TCR) at the interfaces of the TPS layers, pose significant challenges for traditional modeling approaches such as finite-element methods (FEM), as well as reduced-order (ROM) and surrogate (SM) modeling techniques.

What is the new problem that we want to solve? Particularly, a critical artifact in transient reduced-order modeling of hypersonic TPS is the presence of TCRs. Due to manufacturing limitations, when two solids are brought together to form an interface, the true solid-to-solid contact area between them is generally a small fraction of the apparent area over which they meet [Minges1970]. The net effect of imperfect contact is the formation of a temperature discontinuity at the interface that is in general challenging to predict, and that significantly affects the heat transfer across the interface. The identification of TPS properties from internal temperature measurements belongs to a class of *inverse heat conduction problem* (IHCP) [Deng2026], which are known to be ill-posed and sensitive to measurement noise, and thus require careful regularization and robust inference techniques for accurate estimation of TCRs. The challenges posed by IHCPs can introduce significant

uncertainties in transient temperature predictions, and thus compromise the reliability of ROMs for design, analysis, and optimization of hypersonic TPS [Fang2023]. Therefore, there is a critical need for *efficient, accurate, and generalizable* modeling approaches of TPS that can infer TCRs, while providing reliable predictions of internal TPS transient temperatures under varying boundary conditions and material properties.

What has been done in the literature for inference and simulation? Attempts to modeling hypersonic TPS with varying TCRs have taken various forms in the literature, ranging from traditional deterministic physics-based modeling approaches, to probabilistic approaches, and more recently, to scientific machine learning (SciML) techniques that leverage the strengths of both physics-based and data-driven modeling paradigms. For instance, in Ref. [Fang2023], a Lattice Boltzmann Method (LBM) is used to model the transient heat transfer and solid-liquid phase change in a hypersonic TPS, where the TCRs are assumed known, and are shown to significantly delay the melting points of phase changing materials. In Ref. [Zhang2024], authors take one step further by treating the TCRs as unknown parameters, and a finite-element thermo-mechanical model of a three-dimensional reusable heat-pipe cooled thermal protection system is developed to perform Levenberg-Marquardt gradient-based inference of internal TCRs using boundary temperature measurements. A similar approach is adopted in Ref. [Brociek2024], where a finite-difference one-dimensional model of a three-layered TPS is developed to solve the forward problem, and a Levenberg-Marquardt for a Tikhonov regularized optimization is used to solve the inverse problem of inferring TCRs from internal temperature measurements.

Probabilistic approaches to solving the inverse problem of inferring TCRs have also been proposed in the literature. For instance, in Ref. [Gao2026], a Bayesian Network (BN) with sequential importance resample (SIR) is leveraged for probabilistic inversion of TCRs, where the BN is implemented by leveraging a finite-element model of the multi-layered TPS for forward simulations. A BN is also leveraged in Ref. [Deng2026] to infer TCRs, where a neural network is used as a surrogate model to predict internal temperature dynamics due to the computational cost of the FEM-based forward model.

The use of scientific machine learning (SciML) techniques for modeling hypersonic TPS with unknown TCRs has limited exploration in the literature.

.... Other X approaches such as those in Refs.[X] ...

While these methods offer a high-fidelity representation of the underlying physics, highly refined FEMs result in large-scale systems that are computationally expensive to solve, and thus are not suitable for design, analysis, and optimization of hypersonic TPS.

Other data-driven approaches such as those in Refs.[X] ..., leverage Bayesian Networks (BNs) to learn mappings from boundary conditions and thermocop

machine learning techniques to learn the underlying dynamics of the TPS system directly from data, while inferring the TCRs. For instance, in Refs. [X]...

While these methods offer computational efficiency, they often require large amounts of data for training, and may not generalize well to unseen conditions, such as varying boundary conditions and material properties. Moreover, they may not provide insights into the underlying physics of the system, which can be critical for design and optimization.

What is the present work doing? This work presents a systematic design, implementation, and benchmarking of the PIRM for modeling hypersonic TPS with unknown TCRs under experimental constraints such as noisy observables and thermocouple placement limitations.

Objectives of the work The objectives of this work are to,

1. Demonstrate the systematic application of PIROM for modeling hypersonic TPS with unknown contact resistances, emphasizing on the simultaneous inference of material properties and hidden dynamics with limited data.
2. Benchmark the performance of PIROM against traditional physics-based modeling approaches (e.g, one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.
3. Provide insights into the optimal design of physics-based models for inference of contact conductances, including the resolution of the physics-based model, noise levels, and constraints with the placement of thermocouples.

2 Modeling of Hypersonic Thermal Protection Systems

The following discussion presents the modeling of the transient heat conduction problem in multi-layered hypersonic thermal protection systems (TPS) with imperfect contact resistances. Moreover, three different physics-based approaches of decreasing fidelity are presented for solving the governing equations: (i) a fine three-dimensional FEM termed the *full-order model* (FOM), (ii) a one-dimensional FEM with linear elements termed the *one-dimensional FEM* (1D-FEM), and (iii) a lumped-mass representation of the TPS termed the *lumped capacitance model* (LCM). The governing equations and their three solutions strategies are presented next.

2.1 Governing Equations

Consider the generic domain $\Omega \subset \mathbb{R}^d$ show in Fig. 1. A heat flux q_b (i.e., Neumann boundary condition) is prescribed on boundary e_q , while a temperature T_b (i.e., Dirichlet boundary condition) is prescribed on boundary e_T , and $\partial\Omega = \cup_{i=1}^N \partial\Omega_i$ where $e_{iq}, e_{iT} \subset \partial\Omega_i$. The transient heat conduction dynamics is governed by the following energy PDE,

$$\rho c_p \partial_t T - \nabla \cdot (\mathbf{k}(T) \nabla T) = 0, \quad x \in \Omega \quad (1a)$$

$$-\mathbf{k}(T) \nabla T \cdot \mathbf{n} = q_b(x, t), \quad x \in e_q \quad (1b)$$

$$T(x, t) = T_b(x, t), \quad x \in e_T \quad (1c)$$

$$T(x, 0) = T_0(x), \quad x \in \Omega \quad (1d)$$

where the density ρ is constant, while the heat capacity c_p and thermal conductivity $\mathbf{k} \in \mathbb{R}^{d \times d}$ are temperature-dependent. The TPS domain is divided into N_C non-overlapping components $\{\Omega_i\}_{i=1}^{N_C}$, where N_I different imperfect contacts are present at the interfaces between components. To aid in subsequent model reduction derivations, each component is further divided into a *bulk subdomain* Ω_i^b and a *contact subdomain* Ω_i^c , where $\Omega_i = \Omega_i^b \cup \Omega_i^c$ and $\Omega_i^b \cap \Omega_i^c = \emptyset$. The bulk subdomains are defined as the regions of the components that are sufficiently far from the interfaces, while the contact subdomains are defined as the regions of the components that are in close proximity to the interfaces. The contact subdomains are introduced to capture the temperature jumps across the interfaces due to the presence of TCRs, and to provide a systematic framework for deriving reduced-order models that separate bulk and contact contributions to the overall TPS thermal response.

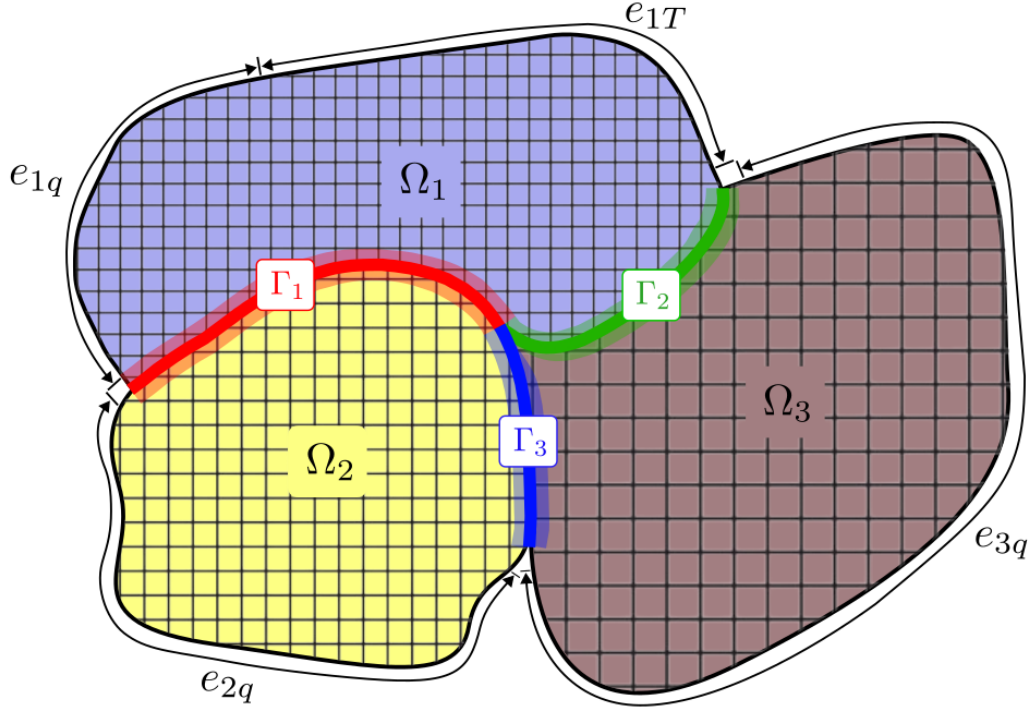


Figure 1: Schematic of the general TPS domain Ω composed of $N_C = 3$ components and $N_\Gamma = 3$ imperfect contacts between components.

For example, Fig. 1 contains $N_C = 3$ components and $N_\Gamma = 3$ imperfect contacts between components. Imperfect contacts translate into *thermal contact resistances* (TCRs), which lead to temperature discontinuities at the interfaces between TPS layers. Therefore, large uncertainties in TCRs may have a significant impact on the overall heat transfer dynamics of TPS. The following discussion presents the simulation of hypersonic TPS under the presence of TCRs, where three different approaches of decreasing fidelity are presented: a full-order three-dimensional finite-element model (FOM), a one-dimensional FEM approximation, and a modified lumped capacitance model (LCM).

2.2 Full-Order Model

The FOM is obtained by spatially discretizing the governing PDE using a Discontinuous Galerkin (DG) FEM to result in a high-dimensional system of ordinary differential equations (ODEs) that captures the transient temperature dynamics of the TPS with high fidelity. As in previous work [VargasVenegas2026], the DG-FEM is chosen to describe the FOM as it: (i) provides a systematic framework for modeling multi-layered TPS with complex geometries and material properties, (ii) provides a natural framework for incorporating TCRs at the interfaces, and (iii) is amenable to systematic model reduction via coarse-graining. In Sec. [X] the FOM numerical solutions are performed using standard FEM instead, and the equivalence between DG and standard FEM is noted upon their convergence.

Consider a DG-FEM spatial discretization of the governing PDE, where the entire domain Ω is

partitioned into M non-overlapping elements $\{E_i\}_{i=1}^M$, where each element belong to one and only one component of the TPS. Moreover, the perfect, imperfect, Neumann, and Dirichlet boundary interfaces are denoted as e_{ij} , Γ_{ij} , e_{iq} , and e_{iT} , respectively. Lastly, $|\partial E_i|$ denotes the length ($d = 2$) or area ($d = 3$) of a boundary ∂E_i . Now, for the i -th element, use a set of P basis functions, such as polynomials, to represent the temperature field as,

$$T_i(x, t) = \sum_{p=1}^P \phi_p(x) u_{i,p}(t), \quad x \in E_i \quad (2)$$

Without loss of generality, let the basis functions on element E_i satisfy the following orthogonality condition,

$$\int_{E_i} \phi_p^i \phi_q^i d\Omega = |E_i| \delta_{pq}, \quad p \neq q \quad (3)$$

where δ_{pq} is the Kronecker delta. Furthermore, to aid in subsequent model-order reduction derivations, choose $\phi_1^i = 1$; therefore, by orthogonality, $\int_{E_i} \phi_p^i d\Omega = 0$ for $p > 1$. The first basis function therefore captures the volume-averaged temperature of the element, while the remaining basis functions capture the spatially-varying temperature distributions within the element.

By standard variational procedure, e.g., [Pernet2018], the full governing equation is denoted as,

$$\mathbf{A}(\mathbf{u}) \dot{\mathbf{u}} = \mathbf{B}(\mathbf{u}) \mathbf{u} + \mathbf{f}^{(b)}(t) + \mathbf{f}^{(c)}(\mathbf{u}, t) \quad (4)$$

where $\mathbf{u} = [\mathbf{u}_1^\top, \mathbf{u}_2^\top, \dots, \mathbf{u}_M^\top]^\top \in \mathbb{R}^{MP}$ corresponds to the DG-FEM states, $\mathbf{A}$ and $\mathbf{B}$ are the mass and stiffness matrices, respectively, which are functions of the temperature-dependent material properties, and $\mathbf{f}^{(b)}$ and $\mathbf{f}^{(c)}$ correspond to the contributions of the boundary conditions and the TCRs, respectively.

2.3 One-Dimensional Finite-Element Model

This approach follows a *one-dimensional* FEM (1D-FEM) approach, where the governing PDE is spatially discretized using a standard FEM with linear basis functions along the thickness direction of the TPS, while the in-plane temperature gradients are neglected. This simplification is commonly used in the literature for modeling heat conduction in multi-layered TPS, since under hypersonic thermal loading, the temperature gradient along the thickness direction of the structure is significantly higher than the in-plane temperature gradients, and thus the through-thickness temperature distribution dominates the overall thermal response of the system [Gao2026].

As with the FOM, the TCR introduces additional considerations in the formulation of the 1D-FEM.

3 Lumped Capacitance Model with Imperfect Contacts

This work leverages the *lumped capacitance model* (LCM), a first-principle physics-based model of the TPS that captures the transient component-wise average temperature of the TPS [Incropera2011]. Nonetheless, the presence of TCRs at the interfaces introduces additional modeling considerations since the temperature drop across an interface includes, in addition to the bulk drop across the interface, a contribution from the contact resistance. The following discussion presents the formulation of the LCM for modeling the transient thermal response of the TPS with unknown TCRs, and the separation of dynamics into *bulk* and *contact* contributions.

Bulk Dynamics Leveraging our previous work in Ref. [VargasVenegas2026], a modified version of the LCM is leveraged in this work as the physics-based component in the PIROM. The LCM may be derived at the component level from a point of view of energy conservation, and is written as,

$$\int_{\Omega_i^b} \rho c_{p,i}(\bar{u}) \dot{\bar{u}}_i d\Omega = \sum_{j \in \mathcal{N}_i^b} \int_{e_{ij}} q_{ij}^b d\Gamma + \sum_{j \in \mathcal{N}_i^c} \int_{\Gamma_{ij}} q_{ij}^c d\Gamma + \int_{e_{iq}} q_b d\Gamma + \int_{e_{iT}} q_T d\Gamma \quad (5)$$

where q_{ij}^b and q_{ij}^c correspond to the heat flux due to neighboring elements with perfect (e_{ij}) and imperfect (Γ_{ij}) interfacing surfaces, while q_{ib} and q_{iT} are the heat flux contributions from the Neumann and Dirichlet boundary conditions on the e_{iq} and e_{iT} interfaces. A multi-layered example is provided in Appendix [X] describing the derivation of eq. (5) from a coarse DG-FEM discretization of the governing PDE.

Closure for Interfacial Heat Flux The presence of TCRs at the component interfaces introduces additional modeling considerations for the heat flux constraints q_{ij}^c . Consider Fig. [X], where the interface $\Gamma_r = \partial\Omega_r^{c,-} \cap \partial\Omega_r^{c,+}$ is magnified to depict the governing heat transfer mechanisms, and where $\Omega_r^{c,-}$ and $\Omega_r^{c,+}$ denote the left and right *contact subdomains* of components that share an imperfect interface Γ_r , respectively. Moreover, let $\bar{\theta}_r^-$ and $\bar{\theta}_r^+$ denote the average trace temperatures to the left and right of the interfaces defined as,

$$\bar{\theta}_r^- = \frac{1}{|\Gamma_r|} \int_{\Gamma_r} T^- d\Gamma, \quad \bar{\theta}_r^+ = \frac{1}{|\Gamma_r|} \int_{\Gamma_r} T^+ d\Gamma \quad (6)$$

where T^- and T^+ are the temperature fields in $\Omega_r^{c,-}$ and $\Omega_r^{c,+}$, respectively. Now, the total temperature difference between the lumped masses across the interface Γ_r is given by,

$$\underbrace{\bar{u}^- - \bar{u}^+}_{\text{total lumped drop}} = \underbrace{(\bar{u}^- - \bar{\theta}_r^-) + (\bar{\theta}_r^+ - \bar{u}^+)}_{\text{bulk drop}} + \underbrace{(\bar{\theta}_r^- - \bar{\theta}_r^+)}_{\text{contact drop}} \quad (7a)$$

$$= (\bar{u}^- - \bar{\theta}_r^-) + (\bar{\theta}_r^+ - \bar{u}^+) - \Delta\bar{\theta}_r \quad (7b)$$

Additionally, define the bulk thermal resistances of the left and right layers as $R_r^- = l_r^- / k_r^- (\bar{u}^-) A$ and $R_r^+ = l_r^+ / k_r^+ (\bar{u}^+) A$, respectively, and let $S_r(\bar{\mathbf{u}}) = R_r^- + R_r^+$ be the effective bulk resistance between the lumped masses. Moreover, the contact resistance and conductances across the interface are denoted as R_r and $g_r = 1/R_r$, respectively. By conservation of energy, the total bulk heat flux across the interface is equal to the contact heat flux across the interface, therefore,

$$\bar{q}_r(t) = \frac{\bar{u}^- - \bar{u}^+}{S_r(\bar{\mathbf{u}}) + R_r} = \frac{(\bar{u}^- - \bar{u}^+) - \Delta\bar{\theta}_r}{S_r(\bar{\mathbf{u}})} = g_r \Delta\bar{\theta}_r \quad (8)$$

Temperature Jump Dynamics Under the presence of TCRs, the classical LCM is modified to include the contact temperature jumps $\Delta\bar{\theta}_r$ as additional states, governed by the heat-flux continuity constraints across the interfaces. The temperature jump dynamics is derived from either: (i) the DG-FEM formulation using a coarse-grained approach as detailed in [APPENDIX], or (ii) the energy conservation constraints at the interface level. From the energy conservation point of view, the algebraic constraints in Eq. [X] are differentiated with respect to time to capture the transient response of the contact temperature jump to changes in boundary conditions, material properties, and TCRs.

For instance, the heat flux closure in Eq. (8) is differentiated with respect to time to yield,

$$(1 + g_r S_r(\bar{\mathbf{u}})) \Delta \dot{\theta}_r = (\dot{u}^- - \dot{u}^+) - \frac{(\bar{u}^- - \bar{u}^+) - \Delta \bar{\theta}_r}{S_r(\bar{\mathbf{u}})} \dot{S}_r(\bar{\mathbf{u}}), \quad r = 1, 2, \dots, N_\Gamma \quad (9)$$

Collecting terms for all interfaces yields the following system of ODEs governing the dynamics of the contact temperature jumps,

$$\mathbf{C}(\bar{\mathbf{u}}; \mathbf{R}_x) \quad (10)$$

$$\Delta \dot{\theta}_r = \frac{(\dot{u}^- - \dot{u}^+) - g_r \Delta \bar{\theta}_r \dot{S}_r(\bar{\mathbf{u}})}{1 + g_r S_r(\bar{\mathbf{u}})} \quad (11)$$

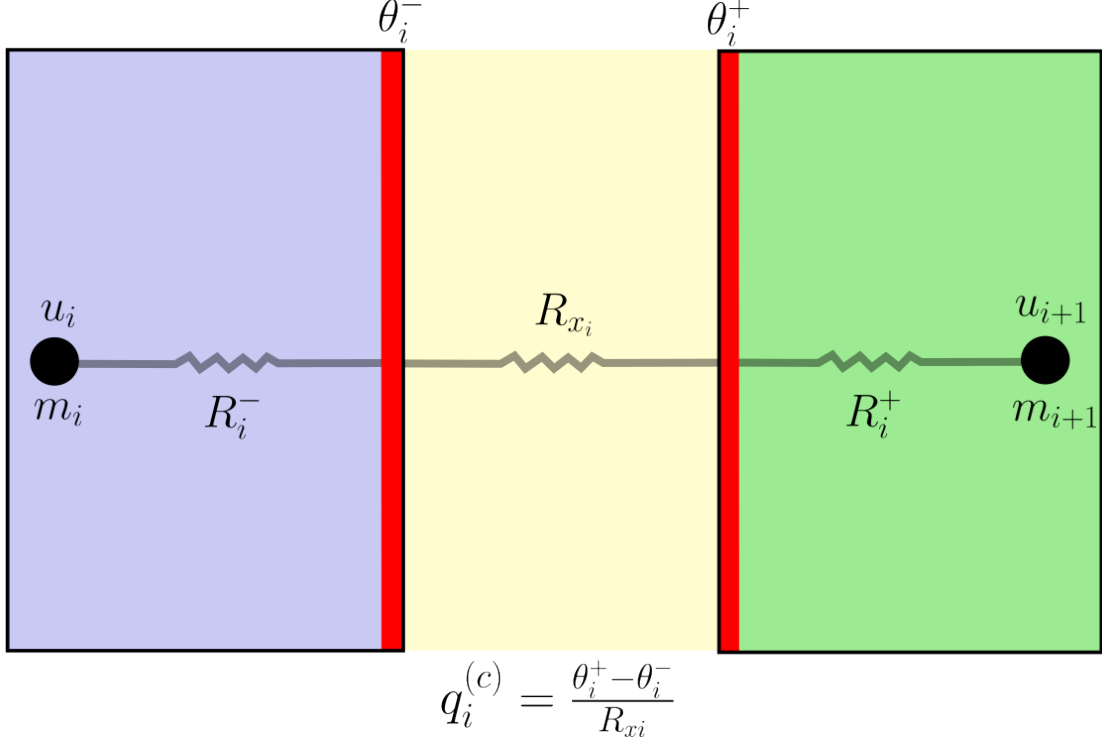


Figure 2: Schematic of the interface between two layers of the TPS, where the temperature drop across the interface is decomposed into a bulk drop and a contact drop.

$$\frac{(\bar{u}_i - \bar{u}_j) - \Delta \bar{\theta}_r}{S_r(\bar{\mathbf{u}})} - g_r \Delta \bar{\theta}_r = 0 \quad (12)$$

Therefore, at an interface $i \in \mathcal{I}$ with a contact resistance, the heat flux between neighboring components may be written as,

$$q_{ij} = G_{ij}(\bar{\mathbf{u}}) \left((\bar{u}_i - \bar{u}_j) - \Delta \theta_i \right) \quad (13)$$

Consider the lumped-mass average temperature state,

$$\mathbf{u} = [u_1, u_2, u_3, u_4]^T \in \mathbb{R}^N \quad (14)$$

and temperature-dependent lumped-capacitance matrix,

$$A(\mathbf{u}) = \begin{bmatrix} a_1(u_1) & 0 & 0 & 0 \\ 0 & a_2(u_2) & 0 & 0 \\ 0 & 0 & a_3(u_3) & 0 \\ 0 & 0 & 0 & a_4(u_4) \end{bmatrix} \quad (15)$$

Let the external Neumann forcing be applied to the first lumped mass,

$$\mathbf{f}(t) = \begin{bmatrix} f(t) \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (16)$$

For interfaces $i = 1, 2, 3$, define the known *bulk resistances*

$$S_i(\mathbf{u}) = R_i^-(\mathbf{u}) + R_i^+(\mathbf{u}) \quad (17)$$

and the unknown contact resistance,

$$R_{xi} > 0, \quad g_{xi} := \frac{1}{R_{xi}}. \quad (18)$$

The classical lumped-capacitance model with effective series resistances is,

$$A(\mathbf{u}) \dot{\mathbf{u}} = B(\mathbf{u}) \mathbf{u} + \mathbf{f}(t) \quad (19)$$

where,

$$B(\mathbf{u}) = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u}) + R_{xi}} \mathbf{K}_i \quad (20)$$

with standard nearest-neighbor diffusion stencils,

$$\mathbf{K}_1 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{K}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad (21)$$

The effective heat flux through interface i is therefore,

$$q_i = \frac{u_{i+1} - u_i}{S_i(\mathbf{u}) + R_{xi}} \quad (22)$$

where the temperature-drop convention is always taken as,

$$\Delta u_i = u_{i+1} - u_i \quad (23)$$

3.1 Total drop = bulk drop + contact drop

To separate bulk and contact contributions, introduce interface temperatures

$$\theta_i^+, \quad \theta_i^-, \quad i = 1, 2, 3, \quad (24)$$

where θ_i^- and θ_i^+ are the temperatures immediately to the left and right of contact i , respectively. Define the contact temperature jumps as,

$$\Delta \theta_i := \theta_i^+ - \theta_i^-, \quad i = 1, 2, 3 \quad (25)$$

The total lumped-to-lumped temperature drop across interface i ,

$$u_{i+1} - u_i \quad (26)$$

This total lumped-to-lumped temperature drop is the sum of the bulk drop across the interface and the contact jump across the interface,

$$\underbrace{u_{i+1} - u_i}_{\text{total drop}} = \underbrace{(\theta_i^- - u_i) + (u_{i+1} - \theta_i^+)}_{\text{bulk drop}} + \underbrace{(\theta_i^+ - \theta_i^-)}_{\text{contact drop}}. \quad (27)$$

Hence,

$$\underbrace{(\theta_i^- - u_i) + (u_{i+1} - \theta_i^+)}_{\text{bulk drop}} = \underbrace{(u_{i+1} - u_i)}_{\text{total drop}} - \underbrace{\Delta\theta_i}_{\text{contact drop}} \quad (28)$$

$$\text{bulk drop across interface } i = (u_{i+1} - u_i) - \Delta\theta_i. \quad (29)$$

Thus the lifted state is

$$x = \begin{bmatrix} u \\ \Delta\theta \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\theta_3 \end{bmatrix} \in \mathbb{R}^7, \quad \Delta\theta = \begin{bmatrix} \Delta\theta_1 \\ \Delta\theta_2 \\ \Delta\theta_3 \end{bmatrix}.$$

3.2 Heat-flux Continuity and DAE Formulation

Using the adopted convention, define the bulk and contact heat fluxes as

$$q_i^{(b)} = \frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)}, \quad q_i^{(c)} = \frac{\Delta\theta_i}{R_{xi}} = g_{xi}\Delta\theta_i. \quad (30)$$

Heat-flux continuity across each interface requires

$$q_i^{(b)} = q_i^{(c)}, \quad i = 1, 2, 3, \quad (31)$$

which yields the algebraic constraints

$$\frac{(u_{i+1} - u_i) - \Delta\theta_i}{S_i(u)} - g_{xi}\Delta\theta_i = 0, \quad i = 1, 2, 3. \quad (32)$$

Lumped-temperature dynamics

The four lumped energy balances yield,

$$a_1(u_1)\dot{u}_1 = f(t) - q_{12}^{(b)} \quad (33a)$$

$$a_2(u_2)\dot{u}_2 = q_{12}^{(b)} - q_{23}^{(b)} \quad (33b)$$

$$a_3(u_3)\dot{u}_3 = q_{23}^{(b)} - q_{34}^{(b)} \quad (33c)$$

$$a_4(u_4)\dot{u}_4 = q_{34}^{(b)} \quad (33d)$$

Substituting using the bulk heat fluxes yields,

$$a_1(u_1)\dot{u}_1 = f(t) - \frac{(u_2 - u_1) - \Delta\theta_1}{S_1(u)} \quad (34a)$$

$$a_2(u_2)\dot{u}_2 = \frac{(u_2 - u_1) - \Delta\theta_1}{S_1(u)} - \frac{(u_3 - u_2) - \Delta\theta_2}{S_2(u)} \quad (34b)$$

$$a_3(u_3)\dot{u}_3 = \frac{(u_3 - u_2) - \Delta\theta_2}{S_2(u)} - \frac{(u_4 - u_3) - \Delta\theta_3}{S_3(u)} \quad (34c)$$

$$a_4(u_4)\dot{u}_4 = \frac{(u_4 - u_3) - \Delta\theta_3}{S_3(u)} \quad (34d)$$

Matrix form of DAE

Define the difference matrix,

$$\mathbf{D} = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \mathbf{D}u = \begin{bmatrix} u_2 - u_1 \\ u_3 - u_2 \\ u_4 - u_3 \end{bmatrix} \quad (35)$$

Define the bulk temperature stencils,

$$\mathbf{K}_1 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{K}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{K}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \quad (36)$$

and the bulk jump-coupling stencils,

$$\mathbf{L}_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{L}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \mathbf{L}_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix} \quad (37)$$

Then,

$$A(u)\dot{u} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} (\mathbf{K}_i u + \mathbf{L}_i \Delta \boldsymbol{\theta}) + \mathbf{f}(t) \quad (38)$$

and the algebraic constraints for heat flux continuity are written as,

$$\text{diag}\left(\frac{1}{S_1(u)}, \frac{1}{S_2(u)}, \frac{1}{S_3(u)}\right) \mathbf{D}u - \left[\text{diag}\left(\frac{1}{S_1(u)}, \frac{1}{S_2(u)}, \frac{1}{S_3(u)}\right) + G_x\right] \Delta \boldsymbol{\theta} = 0 \quad (39)$$

where

$$G_x = \begin{bmatrix} g_{x1} & 0 & 0 \\ 0 & g_{x2} & 0 \\ 0 & 0 & g_{x3} \end{bmatrix} \quad (40)$$

Equation (39), along with the lumped-temperature dynamics, defines a DAE system for the lifted state $x = [u, \Delta \boldsymbol{\theta}]$.

3.3 DAE index reduction

The DAE is index-reduced by differentiating the heat-flux continuity constraints with respect to time, which, for each interface i , yields,

$$\frac{d}{dt} \left[\frac{(u_{i+1} - u_i) - \Delta \theta_i}{S_i(u)} - g_{xi} \Delta \theta_i \right] = 0 \quad (41)$$

The contact conductances g_{xi} are assumed constant, therefore,

$$(1 + g_{xi} S_i(u)) \Delta \dot{\theta}_i = (\dot{u}_{i+1} - \dot{u}_i) - \frac{(u_{i+1} - u_i) - \Delta \theta_i}{S_i(u)} \dot{S}_i(u) \quad (42)$$

Using the original algebraic constraint

$$\frac{(u_{i+1} - u_i) - \Delta \theta_i}{S_i(u)} = g_{xi} \Delta \theta_i, \quad (43)$$

the index-reduced dynamics become,

$$\Delta\dot{\theta}_i = \frac{\dot{u}_{i+1} - \dot{u}_i - g_{xi}\Delta\theta_i\dot{S}_i(u)}{1 + g_{xi}S_i(u)}, \quad i = 1, 2, 3 \quad (44)$$

Define

$$M(u; R_x) = \begin{bmatrix} 1 + g_{x1}S_1(u) & 0 & 0 \\ 0 & 1 + g_{x2}S_2(u) & 0 \\ 0 & 0 & 1 + g_{x3}S_3(u) \end{bmatrix} \quad (45)$$

and,

$$\dot{S}(u) = \begin{bmatrix} \dot{S}_1(u) \\ \dot{S}_2(u) \\ \dot{S}_3(u) \end{bmatrix}, \quad \Delta\theta \odot \dot{S}(u) = \begin{bmatrix} \Delta\theta_1\dot{S}_1(u) \\ \Delta\theta_2\dot{S}_2(u) \\ \Delta\theta_3\dot{S}_3(u) \end{bmatrix} \quad (46)$$

Then, the index-reduced dynamics for the contact jumps can be written in matrix form as,

$$M(u; R_x)\Delta\dot{\theta} = D\dot{u} - G_x(\Delta\theta \odot \dot{S}(u)) \quad (47)$$

Thus, the index-reduced lifted system is given by the coupled ODEs,

$$A(u)\dot{u} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} (\mathbf{K}_i u + \mathbf{L}_i \Delta\theta) + \mathbf{f}(t) \quad (48a)$$

$$M(u; R_x)\Delta\dot{\theta} = \mathbf{D}\dot{u} - G_x(\Delta\theta \odot \dot{S}(u)) \quad (48b)$$

4 Physics-Infused Reduced-Order Modeling

The formulation of PIROM for TPS starts by mathematically connecting the FOM, i.e., the DG-FEM, to the RPM, i.e., the LCM. This results presented in this section leverage the previous work in Ref. X, where a coarse-graining procedure is developed to systematically derive the LCM from the DG-FEM. Such coarse-graining procedure pinpoints the missing dynamics in the LCM when compared to DG-FEM, enabling the systematic design of a memory closure to account for residual dynamics associated with discarded spatial resolution.

4.1 Deriving the Reduced-Physics Model via Coarse-Graining

As demonstrated in previous work [VargasVenegas2026], the LCM can be viewed as a special case of the DG-FEM, where the mesh partition is extremely coarse, and the trial and test functions are piece-wise constants. In this setting, the LCM becomes a coarse zeroth-order DG-FEM with the inverse of thermal resistance chosen as the element-wise penalty factors. In this work, the presence of unknown TCRs motivates the design of a coarse-graining procedure that explicitly accounts for the presence of interfaces with unknown contact conductances, which are not captured by the LCM.

4.1.1 Coarse-Graining of States

Consider a DG-FEM as in Eq. [X] for M elements, and a LCM as in Eq. [X] for N components with N_Γ imperfect interfaces; clearly, $M \gg N$. Define the sets of indices $\mathcal{V}_j$, $\mathcal{F}_i^-$, and $\mathcal{F}_i^+$, corresponding

to the elements contained in component j , and the elements adjacent to interface i on the negative and positive sides, respectively. Formally, the sets are defined as,

$$\mathcal{V}_j = \{i : E_i \subset \Omega_j\} \quad (49a)$$

$$\mathcal{F}_i^- = \{j \in \{1, 2, \dots, M\} \mid \Gamma_i \cap \partial E_j \neq \emptyset, E_j \subset \Omega_i^-\} \quad (49b)$$

$$\mathcal{F}_i^+ = \{j \in \{1, 2, \dots, M\} \mid \Gamma_i \cap \partial E_j \neq \emptyset, E_j \subset \Omega_i^+\} \quad (49c)$$

where $\Gamma_r = \partial\Omega_i^- \cap \partial\Omega_i^+$ is the imperfect interface between the minus Ω_i^- and plus Ω_i^+ components. To aid in the subsequent derivations, introduce a new variable $\mathbf{w} = [\mathbf{w}_1^\top, \mathbf{w}_2^\top, \dots, \mathbf{w}_K^\top]^\top = [\mathbf{w}^-, \mathbf{w}^+]^\top \in \mathbb{R}^{KP}$, where $\mathbf{w}_i$ corresponds to the DG-FEM states of element i , and $\mathbf{w}^- \in \mathbb{R}^{K^-P}$ and $\mathbf{w}^+ \in \mathbb{R}^{K^+P}$ correspond to the DG-FEM states of the elements adjacent to the imperfect contact interfaces on the negative and positive sides, respectively. Note that $K = K^- + K^+$, where K^- and K^+ are the number of elements adjacent to the imperfect contact interfaces on the negative and positive sides, respectively. Clearly, $M > K$, since the total number of elements is much larger than the number of elements adjacent to the imperfect contact interfaces.

Therefore, the volume and left-and-right interface averages for component j and interface r are given by,

$$\bar{u}_j = \frac{1}{|\Omega_j|} \sum_{i \in \mathcal{V}_j} \int_{E_i} \phi^\top \mathbf{u}_i d\Omega_i = \frac{1}{|\Omega_j|} \sum_{i \in \mathcal{V}_j} |E_i| \varphi_i^{j\top} \mathbf{u}_i, \quad j = 1, \dots, N \quad (50a)$$

$$\bar{\theta}_r^-(t) = \frac{1}{|\Gamma_r|} \sum_{m \in \mathcal{F}_r^-} \int_{\Gamma_{rm}} \phi_m^\top \mathbf{w}_m d\Gamma_r = \sum_{m \in \mathcal{F}_r^-} \alpha_{mr}^- \boldsymbol{\mu}_{mr}^\top \mathbf{w}_{mr}^-, \quad r = 1, \dots, N_\Gamma \quad (50b)$$

$$\bar{\theta}_r^+(t) = \frac{1}{|\Gamma_r|} \sum_{m \in \mathcal{F}_r^+} \int_{\Gamma_{rm}} \phi_m^\top \mathbf{w}_m d\Gamma_r = \sum_{m \in \mathcal{F}_r^+} \alpha_{mr}^+ \boldsymbol{\mu}_{mr}^\top \mathbf{w}_{mr}^+, \quad r = 1, \dots, N_\Gamma \quad (50c)$$

where the orthogonal basis functions are defined as $\varphi_i^j = [1, 0, \dots, 0] \in \mathbb{R}^P$. Moreover, $\alpha_{mr}^- = \frac{|\Gamma_{mr}|}{|\Gamma_r|}$ and $\alpha_{mr}^+ = \frac{|\Gamma_{mr}|}{|\Gamma_r|}$ correspond to the fractions of the interface segment Γ_{mr} with respect to the total interface Γ_r , $\mathbf{w}_{mr}^-$ and $\mathbf{w}_{mr}^+$ correspond to the DG-FEM states of the element E_m , which is adjacent to the imperfect interface Γ_r with $r = 1, 2, \dots, N_\Gamma$ on the negative and positive sides, respectively, and $\boldsymbol{\mu}_{mr} = \frac{1}{|\Gamma_{mr}|} \int_{\Gamma_{mr}} \phi_m^\top d\Gamma_{mr} \in \mathbb{R}^P$. Moreover, the $\Gamma_{rm} = \Gamma_r \cap \partial E_m$, and $|\Gamma_{rm}|$ is the point ($d = 2$) or area ($d = 3$) of the interface segment Γ_{rm} .

Conversely, given the average temperatures of the N_C components, as well as the average minus-and-plus temperatures on the two sides of the N_Γ different imperfect interfaces, the DG-FEM states may be written as,

$$\mathbf{u}_i = \sum_{k=1}^{N_C} \varphi_i^k \bar{u}_k + \delta \mathbf{u}_i, \quad i = 1, 2, \dots, M \quad (51a)$$

$$\mathbf{w}_{ir}^- = \sum_{k=1}^{N_\Gamma} \psi_k^{r-} \bar{\theta}_k^- + \delta \mathbf{w}_{ir}^-, \quad i \in \mathcal{F}_r^-, r = 1, 2, \dots, N_\Gamma \quad (51b)$$

$$\mathbf{w}_{ir}^+ = \sum_{k=1}^{N_\Gamma} \psi_k^{r+} \bar{\theta}_k^+ + \delta \mathbf{w}_{ir}^+, \quad i \in \mathcal{F}_r^+, r = 1, 2, \dots, N_\Gamma \quad (51c)$$

where $\psi_k^{r\pm} = \mathbf{0} \in \mathbb{R}^P$ if $k \neq r$, so that the average minus-and-plus temperatures on the two sides of the imperfect contact interface Γ_r only contribute to the states of the elements adjacent to such

interface, and $\delta \mathbf{u}_i$, $\delta \mathbf{w}_{ir}^-$, and $\delta \mathbf{w}_{ir}^+$ correspond to the residuals between the DG-FEM states and the reconstructed states from the average temperatures.

Collectively, Eq. (51c) can be written in matrix form as,

$$\mathbf{u} = \Phi \bar{\mathbf{u}} + \delta \mathbf{u}, \quad \bar{\mathbf{u}} = \Phi^\dagger \mathbf{u} \quad (52a)$$

$$\mathbf{w} = \Pi \bar{\boldsymbol{\theta}} + \delta \mathbf{w}, \quad \bar{\boldsymbol{\theta}} = \Pi^\dagger \mathbf{w} \quad (52b)$$

where $\Phi \in \mathbb{R}^{MP \times NC}$ is a matrix of $M \times N_C$ blocks, with the (i, j) -th block as φ_i^j , $\Phi^\dagger \in \mathbb{R}^{NC \times MP}$ is the left inverse of Φ , with the (i, j) -th block as $\varphi_i^{j\dagger} = \frac{|E_i|}{|\Omega_j|} \varphi_i^{jT}$ and $\delta \mathbf{u}$ is the collection of deviations from the average temperature. Moreover,

$$\Pi = \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{N} \end{bmatrix}, \quad \Pi^\dagger = \begin{bmatrix} \mathbf{M}^\dagger & \mathbf{0} \\ \mathbf{0} & \mathbf{N}^\dagger \end{bmatrix} \quad (53)$$

where $\mathbf{M} \in \mathbb{R}^{K^-P \times N_r}$, $\mathbf{N} \in \mathbb{R}^{K^+P \times N_r}$ correspond to the lifting matrices from the minus-and-plus interface average temperatures $\bar{\boldsymbol{\theta}}^-, \bar{\boldsymbol{\theta}}^+ \in \mathbb{R}^{N_r}$. By their definitions, the projection matrices satisfy,

$$\begin{aligned} \Phi^+ \Phi &= \mathbf{I}, & \Phi^+ \delta \mathbf{u} &= \mathbf{0} \\ \mathbf{M}^+ \mathbf{M} &= \mathbf{I}, & \mathbf{M}^+ \delta \mathbf{w}^- &= \mathbf{0} \\ \mathbf{N}^+ \mathbf{N} &= \mathbf{I}, & \mathbf{N}^+ \delta \mathbf{w}^+ &= \mathbf{0} \end{aligned}$$

4.1.2 Coarse-Graining of Dynamics

Next, the coarse-graining operator for the LCM dynamics is selected as in previous work [VargasVenegas2026], which consists of a Galerkin projection of the DG-FEM dynamics onto the subspace spanned by the coarse-grained states described by the average temperatures. The critical result of the coarse-graining procedure is that the LCM dynamics can be written in the following form,

4.2 Formulation of Reduced-Order Model

The Mori-Zwanzig (MZ) formalism is an operator-projection technique use to derive ROMs for high-dimensional dynamical systems, especially in statistical mechanics and fluid dynamics [Parish2017, Stinis2017, Parish2017]. It provides an exact reformulation of the full-order dynamics in terms of a subset of resolved states. The proposed PIROM is subsequently developed based on such reformulation.

5 Bulk-only Mori–Zwanzig Memory Closure

The purpose of the closure is to account for the unresolved dynamics in the LCM, while leaving the contact-jump dynamics unchanged. Accordingly, the

Accordingly, write the Projected Generalized Langevin Equation (PGLE) for the lumped-temperature dynamics,

$$A(\mathbf{u}) \dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} (\mathbf{K}_i u + \mathbf{L}_i \Delta \boldsymbol{\theta}) + \mathbf{f}(t) + \int_{t_0}^t \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \boldsymbol{\Theta}) ds \quad (55)$$

Now, ensure the designed memory modifies only the bulk conductances, i.e.,

$$\boldsymbol{\kappa} \in \text{span} \{ \mathbf{K}_1 u(t), \mathbf{K}_2 u(t), \mathbf{K}_3 u(t), \mathbf{L}_1 \Delta \boldsymbol{\theta}_1(t), \mathbf{L}_2 \Delta \boldsymbol{\theta}_2(t), \mathbf{L}_3 \Delta \boldsymbol{\theta}_3(t) \} \quad (56)$$

which leads to,

$$\kappa(t, s, u, \mathbf{f}; \Theta) \approx \sum_{j=1}^m \kappa_j(t, s, u, \mathbf{f}; \Theta) \mathcal{P}_j \quad (57)$$

where,

$$\mathcal{P}_j = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} \quad (58)$$

Now, define the memory variables as,

$$\beta_j(t) = \int_{t_0}^t \kappa_j(t, s, u, \mathbf{f}; \Theta) ds \quad (59)$$

then,

$$\int_{t_0}^t \kappa(t, s, u, \mathbf{f}; \Theta) ds = \sum_{j=1}^m \beta_j(t) \mathcal{P}_j \quad (60)$$

It follows that,

$$\sum_{j=1}^m \beta_j(t) \mathcal{P}_j = \sum_{j=1}^m \beta_j(t) \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \left(\sum_{j=1}^m \mathbf{p}_{ij} \beta_j(t) \right) = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{P}_i \boldsymbol{\beta} \quad (61)$$

Then, the memory correction to the LCM dynamics leads to the following modified lumped-temperature dynamics,

$$A(\mathbf{u}) \dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} (\mathbf{K}_i u + \mathbf{L}_i \Delta \boldsymbol{\theta} + \mathbf{P}_i \boldsymbol{\beta}) + \mathbf{f}(t) \quad (62)$$

Now, let,

$$\kappa_i(t, s, \mathbf{u}, \mathbf{f}; \Theta) = e^{-\lambda_j(t-s)} \left(q_j^\top \phi(\mathbf{u}(s)) + r_j^\top \psi(\mathbf{f}(s)) \right) \quad (63)$$

such that the memory variables are given by,

$$\boldsymbol{\beta}(t) = \int_{t_0}^t e^{-\Lambda(t-s)} \left(Q^\top \phi(\mathbf{u}(s)) + R^\top \psi(\mathbf{f}(s)) \right) ds \quad (64)$$

The hidden dynamics are now given by,

$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \quad (65)$$

and in state-space form, the hidden dynamics are,

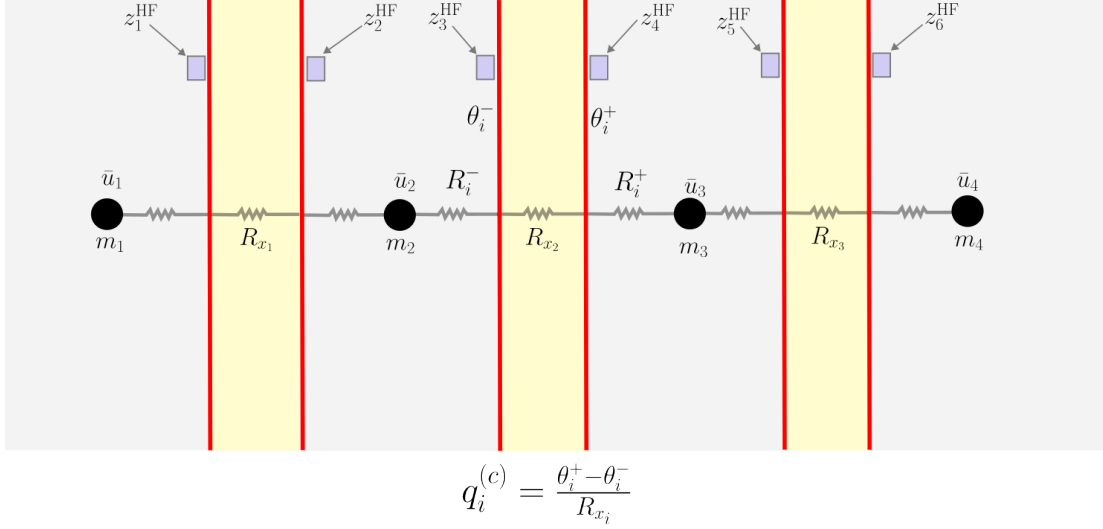
$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \quad (66)$$

The full lifted and closed system is given by,

$$A(\mathbf{u}) \dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} (\mathbf{K}_i \mathbf{u} + \mathbf{L}_i \Delta \boldsymbol{\theta} + \mathbf{P}_i \boldsymbol{\beta}) + \mathbf{f}(t) \quad (67a)$$

$$M(\mathbf{u}; R_x) \dot{\Delta \boldsymbol{\theta}} = \mathbf{D} \dot{\mathbf{u}} - G_x(\Delta \boldsymbol{\theta} \odot \dot{S}(\mathbf{u})) \quad (67b)$$

$$\dot{\boldsymbol{\beta}} = -\Lambda \boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \quad (67c)$$



6 Application to Thermal Protection Systems

In this section, the proposed PIROM approach is applied to modeling a four-layered hypersonic TPS with unknown contact resistances, and is compared against a representative physics-based model, i.e., FEMs, and data-driven surrogate model, i.e., a neural ordinary differential equation (NODE) model. The performance of the models is assessed in terms of their ability to infer the TCRs, and to predict the transient temperature dynamics under varying boundary conditions and material properties. The configuration of the TPS, problem parametrization, optimal model design for inference, and numerical results are presented next.

6.1 Configuration of Thermal Protection System

6.2 Problem Parametrization

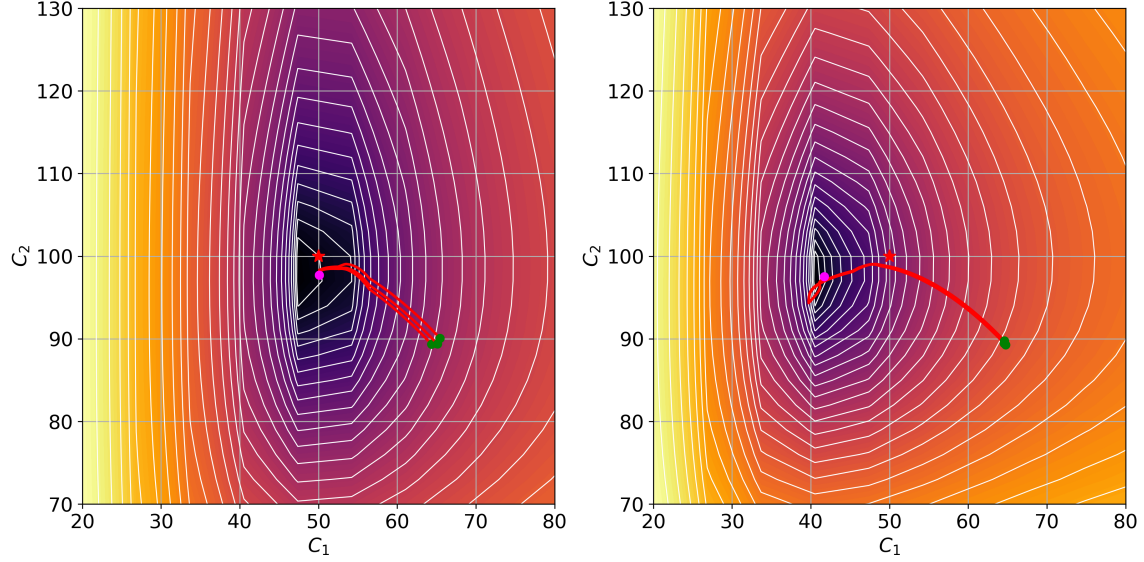
6.3 Optimal Model Design for Inference

This section presents the procedure for optimally selecting the physics-based model for inferring the TCR. Two critical design choices are discussed: (i) formulation of the physics-based model, and (ii) experimental limitations. In the first design choice, the resolution of the physics-based model is selected to balance the trade-off between model bias and computational cost. In the second design choice, experimental limitations, such as the number of thermocouples and their placement, are selected to ensure the identifiability of the TCR inference problem.

6.3.1 Resolution of the Physics-Based Model

The dimensionality of the physics-based model is critical for the performance of the PIROM, as it directly affects the computational cost and identifiability of the TCR inference problem. A coarse model may introduce undesirable model biases that can compromise the accuracy of the inferred TCRs, while a highly-refined model may introduce unnecessary computational costs that hinder efficiency during inference.

One-Dimensional Finite-Element Model



Lumped Capacitance Model

Lumped Capacitance Model with Jump Dynamics

6.3.2 Thermocouple Placement

In addition to the resolution of the physics-based model, the placement of the thermocouples is also critical for the performance of the PIROM, as it directly affects the identifiability of the TCR inference problem.

6.4 Numerical Results

7 Concluding Remarks

The main conclusion of this work are as follows,

1. A one-way physics-data PIROM design is found to successfully infer the unknown TCRs and hidden dynamics simultaneously. During inference, X and Y orders of magnitude in *training efficiency* are obtained relative to the FOM and 1D-FEM forward models, while X and Y orders of magnitude in *evaluation efficiency*

X and Y orders of magnitude in *training efficiency* are obtained relative to the FOM and 1D-FEM based model, while X and Y orders of magnitude in *evaluation efficiency*

successfully infers the unknown TCRs and hidden dynamics simultaneously, offering X and Y orders of magnitude in *training efficiency*, as well as X and Y orders of magnitude in *evaluation efficiency*

relative to the FOM and 1D-FEM based model, while X and Y orders of magnitude in *evaluation efficiency*

and X and Y orders of magnitude in *evaluation efficiency*

TCRs from noisy thermocouple

framework successfully learns a ROM for the prediction of the temperature dynamics of the TPS, while simultaneously inferring the unknown contact conductances with errors of less than X% on noisy data with SNR of up to Y dB.

2. The PIROM outperforms the state-of-the-art data-driven, NN-based SciML, and traditional FEM modeling approaches for inferring contact conductances, as well as for predicting the temperature dynamics for OOD configurations.
3. Model bias is found to significantly affect the

A Appendix

This appendix provides the technical details of the main text.

A.1 Modeling of Hypersonic Thermal Protection Systems

This section provides the detailed formulation of the finite-element model (FEM) based on the Discontinuous Galerkin (DG) method, as well as the one-dimensional FEM, and the lumped capacitance model for the modeling of the thermal dynamics of multi-layered TPS with imperfect interface contacts.

A.1.1 Governing Equations

...

A.1.2 Full-Order Model: Finite Element Method

Domain Discretization Consider the partition of the domain Ω into M non-overlapping elements $\{E_m\}_{m=1}^M$, such that $\cup_{m=1}^M E_m = \Omega$ and $E_m \cap E_n = \emptyset$ for $m \neq n$. For each element, define the set of basis functions $\{\phi_j(x)\}_{j=1}^P$. The temperature field is approximated as,

A.1.3 Weak Form of Discontinuous Galerkin Method

Choosing appropriate basis functions $v(x)$ and $w(x)$, the variational formulation of the governing equations is obtained by applying the DG method element-wise to each component of the TPS. In contrast to the standard perfect-contact formulation, the component interfaces support a finite temperature jump across the interface caused by imperfect thermal contact.

Let $\mathcal{N}_i$ denote the collection of elements neighboring element E_i . The following separation is introduced,

$$\mathcal{N}_i = \mathcal{N}_i^b \cup \mathcal{N}_{ir}^c, \quad i = 1, 2, \dots, M, r = 1, 2, \dots, N_\Gamma \quad (68)$$

where $\mathcal{N}_i^b$ denotes the ordinary neighboring elements coupled by the standard DG interface terms through the interface e_{ij} , and $\mathcal{N}_{ir}^c$ denotes the neighboring elements separated from E_i by an imperfect thermal contact interface $\Gamma_{ir} \subseteq \Gamma_r$, with $r \in \{1, 2, \dots, N_\Gamma\}$ and N_Γ is the total number of imperfect-contact interfaces in the TPS. Formally, the sets are defined as,

$$\mathcal{N}_i^b = \{j \in \{1, \dots, M\} \mid E_j \cap E_i = e_{ij} \neq \emptyset, e_{ij} \subseteq \Omega\}, \quad (69a)$$

$$\mathcal{N}_{ir}^c = \{j \in \{1, \dots, M\} \mid \partial E_i \cap \partial E_j \cap \Gamma_r \neq \emptyset\} \quad (69b)$$

Under this separation, the Interior Penalty Galerkin (IPG) scheme yields the following bi-linear form for the variational formulation of the governing equations [Cohen2018],

$$\sum_{i=1}^M a_{\epsilon,i}(v, w) = \sum_{i=1}^M \left(L_i^{(c)}(v) + L_i^{(b)}(v) \right) \quad (70)$$

where ϵ is a user-specified parameter, and,

$$a_{\epsilon,i}(v, w) = \int_{E_i} \left(\rho_i c_{p,i} v \frac{\partial w}{\partial t} + \mathbf{k}_i \nabla v \cdot \nabla w \right) dE_i - \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} \{ \mathbf{k}_i \nabla v \cdot \mathbf{n}_i \} [w] de_{ij} \\ + \epsilon \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} \{ \mathbf{k}_i \nabla w \cdot \mathbf{n}_i \} [v] de_{ij} + \sigma \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} [v] [w] de_{ij} \quad (71a)$$

$$L_i^{(c)}(v) = \sum_{r=1}^{N_\Gamma} \sum_{l \in \mathcal{N}_{ir}^c} \int_{\Gamma_{il}} v q_{il}^{(c,r)} d\Gamma_{il} \quad (71b)$$

$$L_i^{(b)}(v) = \epsilon \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} (\mathbf{k}_i \nabla w \cdot \mathbf{n}_i) T_b de_{ij} + \int_{e_{iq}} v q_b de_{iq} + \sigma \int_{e_{iT}} v T_b de_{iT} \quad (71c)$$

In the bi-linear form above, the notations of $[]$ and $\{\}$ are respectively the jumps and averages at a boundary e_{ij} shared by two neighboring elements E_i and E_j , defined as,

$$[u] = u|_{E_i} - u|_{E_j}, \quad \{u\} = \frac{1}{2} \left(u|_{E_i} + u|_{E_j} \right), \quad x \in e_{ij} = \partial E_i \cap \partial E_j \quad (72)$$

Furthermore, the terms associated with σ are introduced to enforce the Dirichlet boundary conditions, where σ is a user-specified penalty parameter whose value depend on the size of an element. Depending on the choice of ϵ , the bi-linear form may correspond to symmetric IPG ($\epsilon = -1$), non-symmetric IPG ($\epsilon = 1$), or incomplete IPG ($\epsilon = 0$) schemes. All these schemes are consistent with the original PDE and have similar convergence rate with respect to mesh size.

A.1.4 Discontinuous Galerkin Model

Next, the DG-based FOM is written in an element-wise form. For the i -th element, use a set of P basis functions, such as polynomials, to represent the temperature field as in Eq. [X] that satisfy the conditions in Eq. [X]. Using test functions same as trial functions, the dynamics of the coefficients $\mathbf{u}_i(t) = [u_{i,1}(t), \dots, u_{i,P}(t)]^\top$ for the i -th element is obtained by evaluating the element-wise bi-linear forms,

$$a_{\epsilon,i}(\phi_k^i, T) = L_i^{(c)}(\phi_k^i) + L_i^{(b)}(\phi_k^i), \quad k = 1, \dots, P \quad (73)$$

where ϕ_k^i is the k -th basis function for the i -th element, where the conditions from Eq. [X] are satisfied. Evaluating the bi-linear forms leads to the following system of ODEs,

$$\mathbf{A}_i(\mathbf{u}_i) \mathbf{u}_i = \mathbf{B}_i(\mathbf{u}_i) \mathbf{u}_i + \sum_{j \in \mathcal{N}_i^b} \left(\mathbf{B}_{ij}^i \mathbf{u}_i + \mathbf{B}_{ij}^j \mathbf{u}_j \right) + \mathbf{f}_i^{(c)}(\mathbf{u}, t) + \mathbf{f}_i^{(b)} \quad (74)$$

where, for $k, l = 1, 2, \dots, P$,

$$[\mathbf{A}_i]_{kl} = \int_{E_i} \rho_i c_{p,i} \phi_k^i \phi_l^i dE_i \quad (75a)$$

$$[\mathbf{B}_i]_{kl} = \int_{E_i} \mathbf{k}_i \nabla \phi_k^i \cdot \nabla \phi_l^i dE_i \quad (75b)$$

$$[\mathbf{B}_{ij}^i]_{kl} = \begin{cases} -\int_{e_{ij}} \{\mathbf{k} \nabla \phi_k^i \cdot \mathbf{n}\} \phi_l^i + \sigma [\phi_k^i] \phi_l^i d\Gamma, & j \in \mathcal{N}_i^b \cup \{T\} \\ 0, & \text{otherwise} \end{cases} \quad (75c)$$

$$[\mathbf{B}_{ij}^j]_{kl} = \begin{cases} \int_{e_{ij}} \{\mathbf{k} \nabla \phi_k^j \cdot \mathbf{n}\} \phi_l^j + \sigma [\phi_k^j] \phi_l^j d\Gamma, & j \in \mathcal{N}_i^b \cup \{T\} \\ 0, & \text{otherwise} \end{cases} \quad (75d)$$

$$[\mathbf{f}_i^{(c)}]_k = \begin{cases} \sum_{r=1}^{N_\Gamma} \sum_{l \in \mathcal{N}_{ir}^c} \int_{\Gamma_{il}} \phi_k^i q_{il}^{(c,r)} d\Gamma, & l \in \mathcal{N}_{ir}^c \\ 0, & \text{otherwise} \end{cases} \quad (75e)$$

$$[\mathbf{f}_i^{(b)}]_k = \int_{e_{iq}} \phi_k^i q_b d\mathbf{e}_{iq} \quad (75f)$$

Note that the matrices $\mathbf{A}_i$, $\mathbf{B}_i$, $\mathbf{B}_{ij}^i$, and $\mathbf{B}_{ij}^j$ depend on ρ , c_p , and $\mathbf{k}$, respectively, and hence can be non-linear functions of the state $\mathbf{u}_i$ if the material properties are temperature-dependent. Since the basis functions are orthogonal, if ρc_p is constant within an element, then $\mathbf{A}_i$ is a diagonal matrix; otherwise, $\mathbf{A}_i$ is symmetric and positive-definite, as $\rho c_p > 0$.

For compactness, the element-wise ODEs are stacked together to obtain the following system of ODEs for the entire TPS,

$$\mathbf{A}(\mathbf{u}) \dot{\mathbf{u}} = \mathbf{B}(\mathbf{u}) \mathbf{u} + \mathbf{f}^{(c)}(\mathbf{u}, t) + \mathbf{f}^{(b)} \quad (76)$$

where $\mathbf{u} = [\mathbf{u}_1^\top, \dots, \mathbf{u}_M^\top]^\top \in \mathbb{R}^{MP}$ and $\mathbf{f}^{(b)} = [\mathbf{f}_1^{(b)\top}, \dots, \mathbf{f}_M^{(b)\top}]^\top \in \mathbb{R}^{MP}$, and for $i, j = 1, 2, \dots, M$, it follows that,

$$\mathbf{B}_{ij} = \begin{cases} \mathbf{B}_i(\mathbf{u}) + \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \mathbf{B}_{ij}^i, & i = j \\ \mathbf{B}_{ij}^j, & i \neq j, j \in \mathcal{N}_i^b \cup \{T\} \\ \mathbf{0}, & i \neq j, j \in \mathcal{N}_i^c \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (77a)$$

$$\mathbf{f}_i^{(c)} = \begin{cases} \sum_{r=1}^{N_\Gamma} \sum_{l \in \mathcal{N}_{ir}^c} \mathbf{f}_{il}^{(c,r)}(\mathbf{u}, t), & l \in \mathcal{N}_{ir}^c \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (77b)$$

where $\mathcal{N}_i^c = \bigcup_{r=1}^{N_\Gamma} \mathcal{N}_{ir}^c$. The dependency of $\mathbf{A}$, $\mathbf{B}$, and $q_{il}^{(c,r)}$ on $\mathbf{u}$ is explicitly noted in Eq. [X], which is the main source of non-linearities in the system.

Particularly, the contact heat flux $q_{il}^{(c,r)}$ between elements E_i and E_l across the imperfect-contact interface Γ_r becomes a function of the DG-FEM state $\mathbf{u}$ through the following relationship,

$$\begin{aligned} q_{il}^{(c,r)}(\mathbf{w}, t) &= h_{x,il} (T_i^-(x, t) - T_l^+(x, t)), \\ &= h_{x,il} \left(\sum_{p=1}^P \phi_p^i u_p^i - \sum_{p=1}^P \phi_p^l u_p^l \right), \quad x \in \Gamma_{il} \subseteq \Gamma_r, l \in \mathcal{N}_{ir}^c \end{aligned} \quad (78a)$$

where T_i^- and T_l^+ are the temperature traces on the minus and plus sides of the interface Γ_{il} , respectively, which can be evaluated as a function of the state $\mathbf{u}$ by evaluating the temperature traces on the interfaces.

is written as a function of the state $\mathbf{u}$ by evaluating the temperature traces on the interfaces.

and interface Γ_l is written as a function of the state $\mathbf{u}$ by evaluating the temperature traces on the interface. Let $R_{x,il}$ denote the thermal contact resistance of the interface Γ_{il} , and $h_{x,il} = 1/R_{x,il}$ be the corresponding thermal contact conductance. Moreover, let T_{il}^- and T_{il}^+

Moreover, on the imperfect-contact interface Γ_l , the interface represents a physical thermal resistance between two neighboring components. Let $R_{x,l}$ denote the thermal contact resistance and define the corresponding thermal contact conductance as $h_{x,l} = 1/R_{x,l}$. Then, the heat flux across the interface is governed by,

$$q_{ij}^{(c)} = h_{x,ij} \left(T_i^-(x, t) - T_j^+(x, t) \right), \quad x \in \Gamma_{ij} \quad (79)$$

where T_i^- and T_j^+ denote the temperature traces on the minus and plus sides of the interface Γ_{ij} , respectively. The normal conditions on the interface are given by the conservation of energy across the interface,

$$-\kappa_i \nabla T_i^-(x, t) \cdot \mathbf{n}_i = q_{ij}^{(c)}, \quad x \in \Gamma_{ij}, \quad (80a)$$

$$-\kappa_j \nabla T_j^+(x, t) \cdot \mathbf{n}_j = -q_{ij}^{(c)}, \quad x \in \Gamma_{ij} \quad (80b)$$

where $\mathbf{n}_i = -\mathbf{n}_j$ is the unit normal vector of the interface Γ_{ij} pointing from component i to component j . Thus, across an imperfect contact interface, the heat flux is conserved, while the temperature is allowed to be discontinuous.

A.2 Modified Lumped Capacitance Model

The TPS is divided into N components with M elements. For component i , define Ω_i as the spatial domain of the component, and $\partial\Omega_i$ as the boundary of the component. For each interface between two components, define $l(i)$ and $r(i)$ as the left and right components that form the interface, respectively.

For interface i , define $\Gamma_i = \partial\Omega_{l(i)} \cap \partial\Omega_{r(i)}$,

$$\theta_i^-(t) = \frac{1}{|\Gamma_i|} \int_{\Gamma_i} T_h^-(x, t) d\Gamma, \quad (81a)$$

$$\theta_i^+(t) = \frac{1}{|\Gamma_i|} \int_{\Gamma_i} T_h^+(x, t) d\Gamma \quad (81b)$$

$$\Delta\theta_i(t) = \theta_i^-(t) - \theta_i^+(t) \quad (81c)$$

Define the trace operators,

$$C_i^- \mathbf{u} = \frac{1}{|\Gamma_i|} \int_{\Gamma_i} \sum_{j=1}^{N_u} u_j(t) \phi_j(x) d\Gamma, \quad (82a)$$

$$C_i^+ \mathbf{u} = \frac{1}{|\Gamma_i|} \int_{\Gamma_i} \sum_{j=1}^{N_u} u_j(t) \phi_j(x) d\Gamma \quad (82b)$$

Stacking all the N_Γ interfaces, define the trace extraction operator as,

$$\Psi^+ = \begin{bmatrix} C_1^- - C_1^+ \\ \vdots \\ C_{N_\Gamma}^- - C_{N_\Gamma}^+ \end{bmatrix} \in \mathbb{R}^{N_\Gamma \times MP} \quad (83)$$

where M is the number of elements and P is the number of basis functions per element. Therefore, it follows that,

$$\mathbf{s}(t) = \mathbf{\Psi}^+ \mathbf{u}(t) \quad (84)$$

Thus,

$$\mathbf{\Pi}^+ = \begin{bmatrix} \mathbf{\Phi}^+ \\ \mathbf{\Psi}^+ \end{bmatrix} \quad (85)$$

Introduce the lifting matrix,

$$\mathbf{\Psi} \in \mathbb{R}^{MP \times N_\Gamma} \quad (86)$$

such that,

$$\mathbf{\Phi}^+ \mathbf{\Psi} = \mathbf{0}, \quad \mathbf{\Psi}^+ \mathbf{\Psi} = \mathbf{I} \quad (87)$$

Note that $\mathbf{\Phi} \bar{\mathbf{u}}$ contributes $E_\Gamma \bar{\mathbf{u}}$ to the interface jump, therefore the state decomposition can be written as,

$$\mathbf{u} = \mathbf{\Phi} \bar{\mathbf{u}} + \mathbf{\Psi} (\Delta \boldsymbol{\theta} - E_\Gamma \bar{\mathbf{u}}) + \delta \mathbf{u} = \tilde{\mathbf{\Phi}} \bar{\mathbf{u}} + \mathbf{\Psi} \Delta \boldsymbol{\theta} + \delta \mathbf{u} \quad (88)$$

where $\tilde{\mathbf{\Phi}} = \mathbf{\Phi} - \mathbf{\Psi} E_\Gamma$, and $\delta \mathbf{u}$ is the deviation from the average temperature in the components.

Important point: The critical assumption is that the contact domains $\Omega_r^{c,-}$ and $\Omega_r^{c,+}$ are sufficiently thin such that their transient thermal response is negligible, and thus the contact temperature jumps $\Delta \theta_r$ are governed by algebraic constraints for heat-flux continuity across the interfaces.

A.3 Coarse-Graining Example

Consider the three-layered TPS shown in Fig. [X] with three components $\{\Omega_j\}_{j=1}^3$ and three contact interfaces $\{\Gamma_k\}_{k=1}^3$. The set of elements for each component, and on the minus and plus sides of each interface are defined as,

$$\mathcal{V}_j = \{m \in \{1, \dots, M\} : E_m \subseteq \Omega_j\}, \quad (89a)$$

$$\mathcal{F}_k^- = \{m \in \{1, \dots, M\} : E_m \cap \Gamma_k \neq \emptyset, E_m \subseteq \Omega_{l(k)}\}, \quad (89b)$$

$$\mathcal{F}_k^+ = \{m \in \{1, \dots, M\} : E_m \cap \Gamma_k \neq \emptyset, E_m \subseteq \Omega_{r(k)}\} \quad (89c)$$

where $\Omega_{l(k)}$ and $\Omega_{r(k)}$ are the left and right components that form the interface Γ_k , respectively, with $\Gamma_k = \partial \Omega_{l(k)} \cap \partial \Omega_{r(k)}$.